ClassGuard

App Guide & Project Status

Version 2.3.0: what ClassGuard does, how to use each part, what is built, how it was tested, and what is still pending.

Hikvision · Dahua · CP Plus · UNV · ONVIF · Any RTSP camera · NVR / DVR

iVMS-style Live View · NVR playback · Remote configuration · Face attendance

Class mood alarm · Person tracking · Safety watch

Version	Code	Branch	Date
2.3.0	amit5042k/class-gaurd	claude/camera-management-desktop-app-bR07t	6 November 2026

1 · AT A GLANCE

What is ClassGuard?

A Windows desktop app for schools, working like Hikvision iVMS-4200 and CP Plus KVMS Pro combined. It shows cameras and NVRs of several brands on one screen, records them, plays back what the NVR recorded, configures the devices, and adds AI: face attendance, class mood alarms, person tracking and safety alerts.

Screens	Camera brands	Automated tests	Requested features built
14	6	119	All

Features

Live View
Device tree, drag to windows, 1–36 windows, saved views, auto-switch, digital zoom, PTZ, events.

Playback
Footage on the NVR's own disk, and ClassGuard's recordings on the PC. Timeline, clips, snapshots.

Devices
Add NVRs and cameras, import channels, and change their video settings, time and more remotely.

Face attendance
Enroll photos; automatic check-in, Present / Late / Absent, monthly report.

Class mood alarm
Share of upset faces in a class over the last minute; alarm above a set level.

Person tracking
Type a name: where the student or staff member is now and where they went.

Safety watch
Restricted zones; helmet / mask / vest checks with your own YOLO model.

Users & security
Sign-in, 3 roles, lock screen, audit log, encrypted camera passwords.

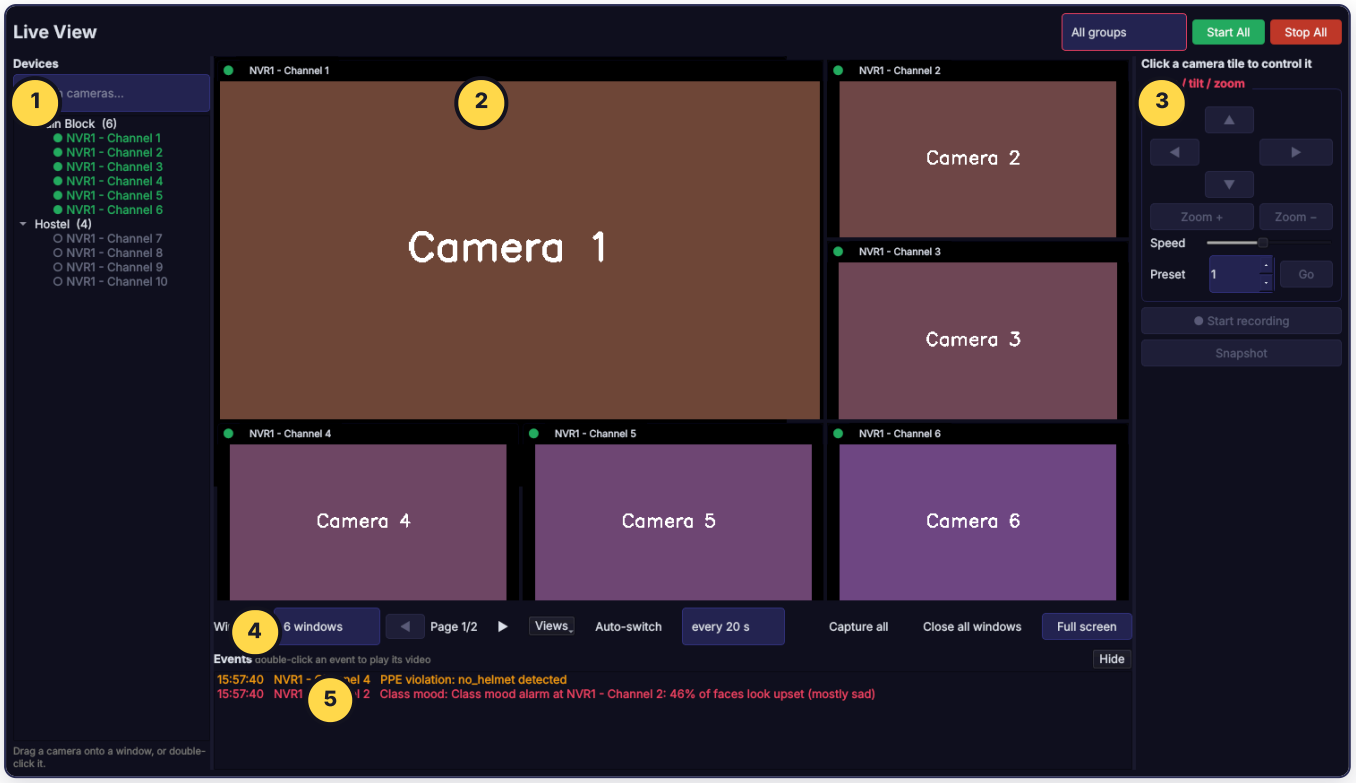
Backup & install
Daily backups, restore, Windows installer, built-in self-test.

Status in one line: everything requested is built and passes 119 automated tests. On your real Hikvision NVR (10.0.5.1), login, channel import, still pictures and connection checks work. Smooth **live video** on every channel, NVR playback and remote configuration still have to be confirmed there (section 8).

2 · LIVE VIEW

Watching cameras

Laid out like desktop VMS software: devices on the left, windows in the middle, controls on the right, events below.



1. **Device list.** Each NVR or group with its channels: green = live, orange = connecting, red = problem. Search box. **Double-click** or **drag** a channel onto a window; double-click an NVR to show all its channels.
2. **Windows.** 1, 4, 6, 8, 9, 16, 25 or 36 (6 and 8 have one large window). Drag a window onto another to swap. Double-click = single view.
3. **Control panel.** Pan / tilt / zoom and presets for PTZ cameras, record, snapshot.
4. **Window toolbar** (on mouse-over): Snapshot, Record, Zoom (mouse wheel), Replay (last 30 s), Close. Labels: **STILL** = still pictures, **CONNECTING**, **⚠ NO FULL PICTURE**.
5. **Bottom bar and events.** Pages ◀ ▶, **Views** (save / open window arrangements), **Auto**-switch every 10 s – 2 min, Capture all, Clear, Full screen (F11). Alerts appear in the event list; double-click one to play its video.

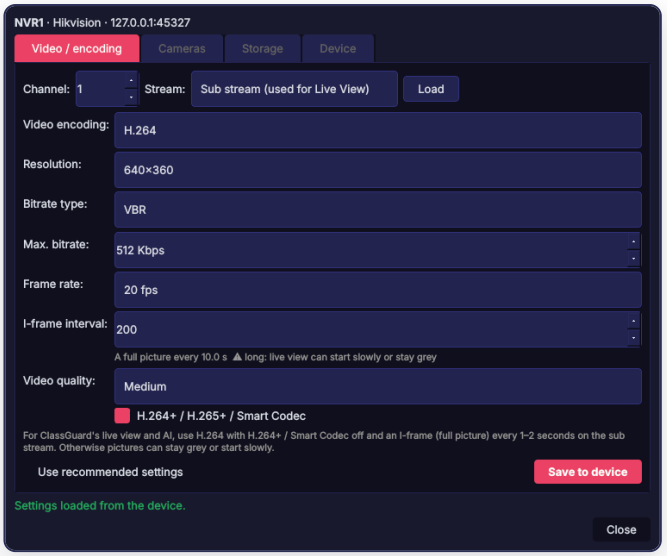
Fast start

Only the cameras on screen stream (plus any that record or run AI), started a few at a time, so a 32-channel NVR is not overloaded. ClassGuard's own **direct decoder** is tried first (first picture in under a second in tests); on Hikvision a still picture from the NVR shows at once while video connects. The decoder that worked is remembered per camera, so next time it connects straight away.

Add and configure NVRs and cameras

Add

1. **NVR / DVR → + Add**, *Test connection*, then **Import channels**: only channels with a camera are added. **Check channels** re-tests them.
2. Or **Camera Management → + Add Camera** (brand, IP, login, channel). *Get from camera (ONVIF)* fills the address in; **ONVIF Discover** scans the network.
3. Per camera: sub-stream, RTSP transport, AI switches, recording mode, PTZ type.



Configure (new)

Configure on an NVR or camera row opens the device's own settings (Hikvision ISAPI; Dahua / CP Plus CGI):

- **Video / encoding** per channel and stream: codec, resolution, bitrate, frame rate, **I-frame interval**, quality, **H.264+ / Smart Codec**. **Use recommended settings** = H.264, H.264+ off, a full picture every 2 s.
- **Cameras**: channel list with IP and Online / OFFLINE.
- **Storage**: disks, capacity, free space, status.
- **Device**: model, serial, firmware, clock, set time from the PC, reboot.

If a camera won't show a picture: read the tile, then Diagnose

Tile says	What to do
Can't reach IP:554	Check IP, cable / Wi-Fi, and that RTSP port 554 is open
Wrong username or password	Fix the login (NVR / DVR → Edit updates every channel)
No video on this channel / sends no video	No camera, or camera offline on that NVR channel (see Configure → Cameras)
STILL · ⚠ NO FULL PICTURE	Configure → channel → Sub stream → **Use recommended settings** → Save
Anything else	Right-click → **Diagnose connection…** → Copy, and send the report to support

Diagnose checks the network, login, raw video data, full pictures (keyframes), the NVR's still picture, and each decoder over TCP and UDP. When nothing decodes it saves a short video sample for support.

4 · PLAYBACK

Recordings on the NVR and on the PC

Recordings has two tabs: what the NVR / camera stored on its own disk, and what ClassGuard recorded on this PC.



Recorded on the NVR / camera (new)

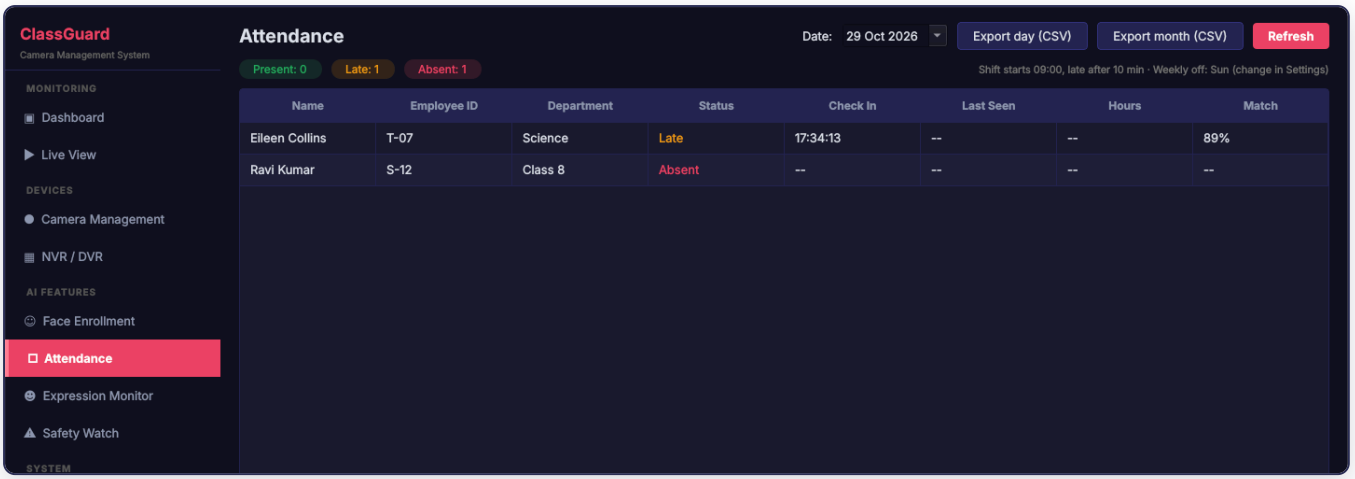
- Choose camera and date: ClassGuard searches the recorder and draws the day: 🟦 continuous, 🟧 motion, 🟥 event.
- Click the bar, or type a time and press **Play**.
- ⏪ **30 s / 30 s** ⏩, Pause, **Snapshot**, **Save clip** (MP4 on the PC).
- Timeline for Hikvision, Dahua and CP Plus; UNV plays from a typed time.
- Live View **Replay** and alert videos use it when the PC has no recording.

Recorded by ClassGuard (this PC)

- Per camera: Off, Continuous or Motion recording, or manual from Live View.
- Player: 0.5× – 8×, ±10 s, frame by frame, snapshot, export.
- Alert → **Video** opens the recording 5 s before the alert.
- Clean-up: keep N days, at most N GB, always leave N GB free.

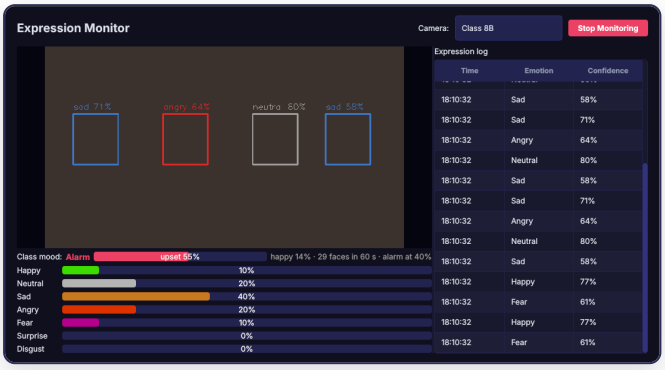
Attendance and class mood

1 · Add person name, ID, class → **2 · Enroll 3–5 photos** webcam or file → **3 · Camera sees them** Attendance switched on → **4 · Check-in saved** with photo



- **Late** = first seen after shift start + grace minutes; **Absent** = not seen on a working day (Settings). **Monthly CSV** with P / L / A / O per day.

Class mood alarm (new)



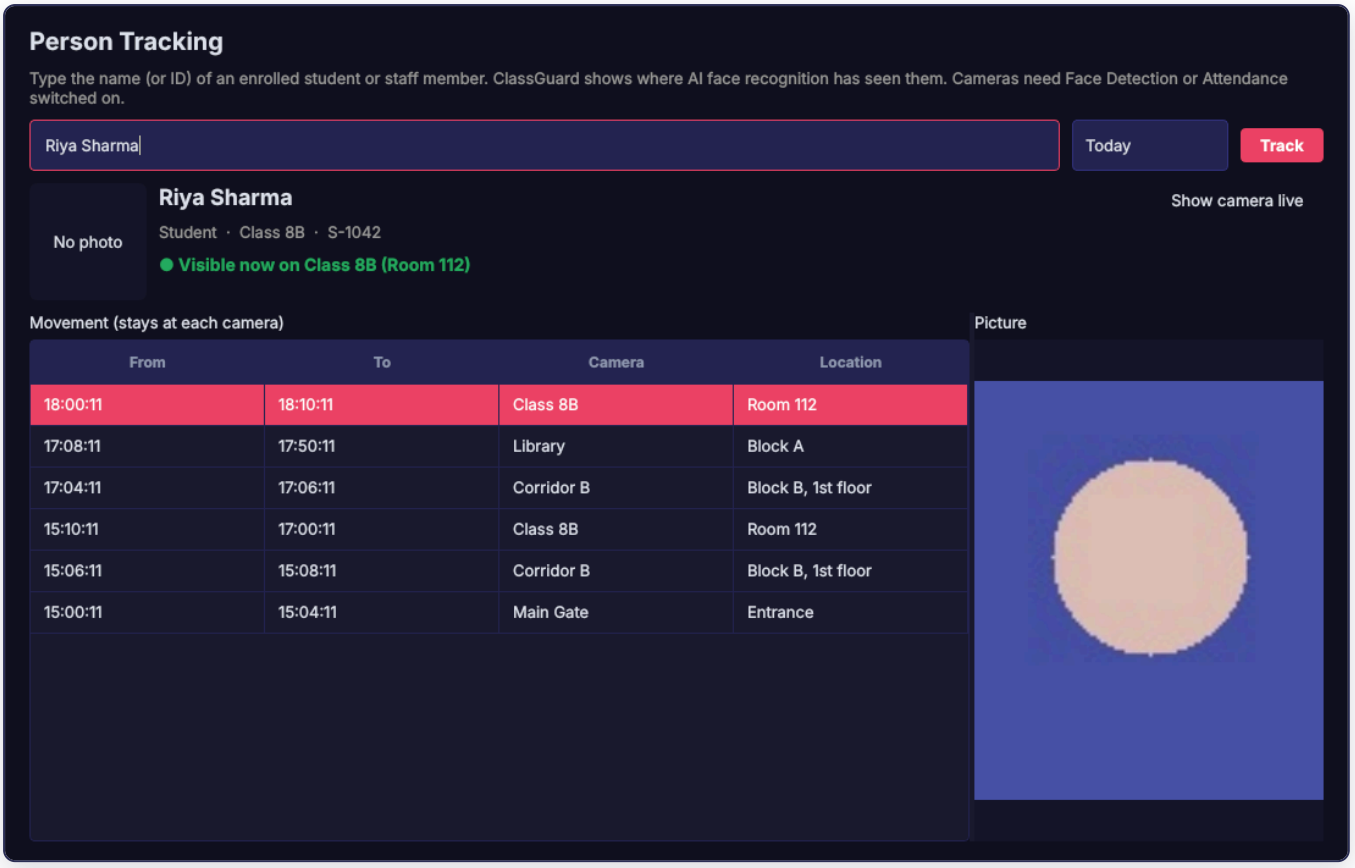
On cameras with **Expression Monitor** switched on, every face is given one of 7 expressions. ClassGuard counts the **upset** ones (angry, fear, disgust, sad) among all faces over the last minute and shows the class mood: **Good**, **Mixed**, **Low** or **Alarm**.

At **Alarm** (default: 40 % upset, at least 8 faces, over 60 s) a high-priority alert is raised with a photo, e.g. *"Class mood alarm at Class 8B: 55 % of faces look upset (mostly sad)"*. All three numbers are in **Settings → Class mood alarm**.

Needs the optional emotion model (DeepFace, about 1 GB). Expressions are an indication, not a diagnosis.

Person tracking and safety

Person tracking (new)



- Open **Person Tracking**, type a name or ID (suggestions appear as you type) and press **Track**.
- Shows **Visible now on …** or **Last seen on … (12 min ago)**, the **movement** from camera to camera with times, and a face picture for each stay. **Show camera live** opens that camera.
- While someone is tracked, a pop-up appears each time they are seen. Range: today, 24 h, 7 or 30 days. History kept 30 days (Settings).
- Works on cameras with Face Detection or Attendance switched on, for enrolled people. Admin and Operator only.

Safety Watch

Draw a restricted zone; anyone whose feet are inside raises an alert with a photo. Helmet / mask / vest checks need your own YOLO model (.onnx or Darknet); none is included.

Privacy

Faces are biometric data. Tell staff, students and parents, get the consent your rules require, limit who can track people, and keep history no longer than needed.

Behind the scenes

Users and audit log

Viewer (watch, play back), Operator (+ PTZ, recording, AI, faces, tracking), Administrator (+ devices, settings, users). 5 wrong passwords block an account for a minute. The audit log records sign-ins, device and setting changes (including remote configuration), exports, deletions and backups.



Backups

Daily automatic backup (last 7 kept), **Back up now** and **Restore** in Settings: database, settings, faces and the encryption key (not recordings).

Where data is kept

On the PC in `C:\Users\<you>\.classguard\`. Camera passwords are encrypted; sign-in passwords stored only as hashes.

How video flows

Camera / NVR
RTSP (TCP / UDP)

→

Stream thread
one per camera; direct → standard → alternative decoder; still pictures as fallback

→

Screen
up to 15 pictures / s

Every frame

→

AI worker
faces, expressions, safety (newest frame only)

→

Attendance · mood · tracking · alerts
database

→

Recorder
MP4 segments, clean-up

Device adapters

`hikvision ISAPI` `dahua / CP Plus CGI` `onvif` `ptz` `rtsp diag` `nvr_playback` `device_config`

Engines

`stream_manager` `rtp_capture` `recording` `ai_worker` `class_mood` `tracking` `attendance` `alerts` `backup`

7 · WHAT'S BUILT

Built features and how each was checked

YOUR NVR confirmed on your Hikvision NVR TEST CAMERA end-to-end with a simulated RTSP camera AUTOMATED TEST automated tests TO CONFIRM still to confirm on real devices

Feature	Since	Verification
NVR add, test, channel import / check	1.x	YOUR NVR AUTOMATED TEST
Login (Digest), stream request, keyframes	2.0	YOUR NVR Diagnose report: data and full pictures arrive
Still pictures from the NVR	2.0.8	YOUR NVR channel 4 showed a picture
Live video, direct decoder, fast start	2.1.2	TEST CAMERA H.264 + H.265 TO CONFIRM on your NVR
iVMS-style Live View (tree, drag, layouts, views, auto-switch, zoom, events)	2.1	AUTOMATED TEST YOUR NVR screen in use
PTZ (Hikvision, Dahua, ONVIF)	1.x	AUTOMATED TEST commands TO CONFIRM
NVR playback (search, timeline, play, clip)	2.2	TEST CAMERA AUTOMATED TEST TO CONFIRM
Remote configuration (video, I-frame, H.264+, time, disks, reboot)	2.3	AUTOMATED TEST simulated Hikvision TO CONFIRM
PC recording (continuous, motion, manual) + player	1.x	TEST CAMERA AUTOMATED TEST
Face enrollment, recognition, attendance	1.x	TEST CAMERA real face photo AUTOMATED TEST
Class mood alarm	2.1	AUTOMATED TEST TO CONFIRM needs emotion model
Person tracking by name	2.1	AUTOMATED TEST TO CONFIRM with enrolled people
Safety zones, alerts, pop-ups	1.x	AUTOMATED TEST
Users, roles, lock, audit, encryption	1.x	AUTOMATED TEST
Backup / restore	1.x	AUTOMATED TEST
Windows installer + self-test	1.x	Built and self-tested on Windows for every version (GitHub Actions); YOUR NVR installed on your PC

Recent versions

Version	Changes
2.0.8	Still pictures when video stays grey; keyframe counting in Diagnose
2.1.0 / 2.1.1	iVMS/KVMS-style Live View, class mood alarm, person tracking; direct decoder; layout fixes
2.1.2	Instant start: direct decoder first, decoders remembered, still picture at once
2.2.0 · 2.3.0	NVR playback · remote NVR and camera configuration

8 · WHAT'S PENDING

Still to do

A · To confirm on your devices (no new code expected)

Item	How to check	Status
Live video on every channel	Install 2.3.0, Start All. If a tile shows STILL / ⚠, run Configure → Use recommended settings, or send its Diagnose report	IN PROGRESS
NVR playback	Recordings → NVR tab → today → click the timeline	TO TEST
Remote configuration	NVR / DVR → Configure: load channel 1 sub stream, save a change	TO TEST
PTZ on a PTZ camera	Set Type: PTZ Camera, move, stop, preset	TO TEST
Attendance, mood, tracking	Enroll 3 people; a classroom camera with Attendance + Expression on	TO TEST
Dahua, CP Plus, UNV, ONVIF devices	One of each if the school has them	TO TEST

B · Not built yet (possible next features)

Feature	Notes
Fast-forward / reverse in NVR playback; download whole NVR files	Playback is normal speed; clips are saved while playing
Remote config for UNV and ONVIF; image (brightness, IR), OSD, motion-detection setup	Video encoding, time, disks and reboot exist for Hikvision, Dahua, CP Plus
E-map (cameras on a floor plan), two-way audio, multiple monitors	Common in iVMS / KVMS
Email / WhatsApp / SMS alerts; mobile or web viewer	Alerts are inside the app only
GPU acceleration for AI on many cameras	AI on 4–6 cameras at once on a normal PC
A trained helmet / vest model; signed installer	Signing (paid certificate) stops Chrome / Windows "dangerous file" warnings

Known limitations

- About 16 live cameras at once on a normal PC; use sub-streams for big grids.
- Emotion analysis needs a separate 1 GB install (DeepFace). Expression and mood results are indications only.
- PC recordings are MP4 (MPEG-4 Part 2): they play in ClassGuard and VLC.

9 · NEXT STEPS

Going live

Step 1
Finish the NVR trial

- Install 2.3.0; in Chrome's download list choose **Download dangerous file** (the installer isn't signed yet).
- NVR / DVR → **Configure** → each channel's Sub stream → **Use recommended settings** → Save.
- Start All; send the Diagnose report of any channel that isn't live video.

Step 2
Pilot 2 weeks

- One building: attendance and mood in two classrooms, tracking at the gate and corridors, one safety zone.
- Tune match threshold, mood alarm level, recording limits; check daily backups.

Step 3
Extras as needed

- Choose from section 8B: alerts by WhatsApp / email, e-map, playback speed, signed installer.

Acceptance checklist

- [ ] All channels show live video within a few seconds
- [ ] Drag, layouts, saved view and auto-switch work
- [ ] NVR playback plays yesterday's recording; clip saved
- [ ] Configure: I-frame change saved and visible on the NVR
- [ ] Device time synced; disk status shown
- [ ] PTZ moves, stops, goes to a preset
- [ ] 16 cameras for 1 hour without slowdown
- [ ] Enrolled person recognised at 1–3 m; check-in saved
- [ ] Late / Absent correct; monthly CSV opens in Excel
- [ ] Class mood shows; alarm fires on a staged test
- [ ] Tracking a name shows "Visible now" and the route
- [ ] Zone intrusion raises an alert with photo and video
- [ ] Viewer account can't change anything
- [ ] Backup restored on a second PC

Summary. All requested features are built: iVMS / KVMS-style live view, NVR playback, remote configuration, face attendance, class mood alarm, person tracking and safety watch, with 119 automated tests. What remains is confirming live video, playback and configuration on your NVR (Step 1), then a short pilot.